

lecture

November 12, 2025

positive result:

assume realizability, assume m combinations of 3 points instead of 4 point comparisons

$O(\sqrt{m})$ suffices

create dependency graph, draw edges for constraint (abc), draw on (ab),(ac)

the answer will depend on *arboricity*

Definition 1 (arboricity). the arboricity of a graph is the min # of forests in which edges can be partitioned

if r is the arboricity, and S is the set of subgraphs of G ,

$$r = \max_S \frac{|E_S|}{|V_S| - 1} \quad (1)$$

if a graph G has arb r , then it is $2r$ colorable

$$r \leq 2\sqrt{|E_G|}$$

$\exists$ an ordering of $V \in G$, $x_1, \dots, x_n$, every $|N(x_i)| \leq 2r - 1$ for $i < j$ for x_j where N gives the neighbors of x